Malaria Disease Image Classification

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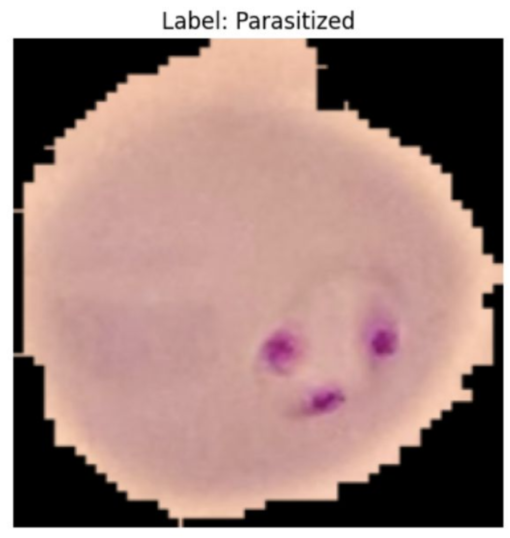
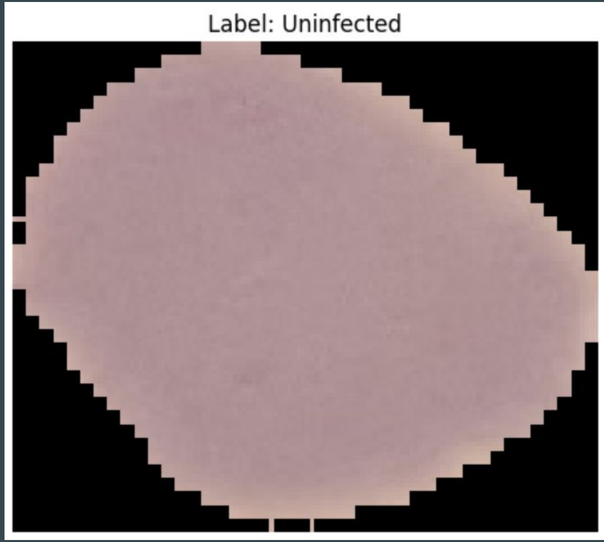
The Problem

- Two sets of images - infected, uninfected
- Using machine learning, train a program in order to differentiate between the two





The Solution

- Using resnet18, classify parasitized and uninfected cells
- Using the classification (binary classification, since it's two categories), be able to use the program to identify cells' state

```python
from sklearn.model_selection import train_test_split

## Images:
from skimage.transform import resize
from skimage.io import imread

import os
import matplotlib.pyplot as plt
import pickle

device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
print("Using device:", device)
```

```
Using device: cuda
```

```python
datadir = '.\\Images\\'
categories = ['Parasitized', 'Uninfected']

X = []
y = []

for i in categories:
    print ('loading... category: ', i)
    path = os.path.join (datadir, i)
    print(path)
    for img in os.listdir (path):
        img_array = imread (os.path.join (path, img))
        img_resized = resize (img_array, (224, 224, 3))
        img_tensor = torch.tensor (img_resized).float().to(device) # convert to tensor and send to device (GPU/CPU)
        X.append(img_tensor) # permute to (C, H, W) format and add batch dimension
        y.append (torch.tensor (categories.index(i)).to(device))
    print ("loaded category ", i, "successfully")

X = torch.stack(X)  # stack all tensors in a single tensor
y = torch.stack(y)  # stack all labels in a single tensor
X.shape, y.shape
```

Results

- After the program runs, an almost perfect accuracy score
- Can be used in real world situations to identify diseases and diagnose people

```
[60]  ✓  0.5s

...  Correct=383, Total=383, Test Accuracy: 100.0000%
     Correct=95, Total=96, Test Accuracy: 98.9583%
```